Introduction to Data Science

(Also available in CODAP)

Students learn about Categorical and Quantitative data, are introduced to Tables by way of the Animals Dataset, and consider what questions can and cannot be answered with available data.

Lesson Goals	Students will be able to… • Explain the difference between Categorical and Quantitative data • Identify whether a variable in a dataset is Categorical or Quantitative • Identify the Header Row and Identifier Column of a Table
Student-facing Lesson Goals	• Let's learn about data inside tables.
Materials	• Lesson Slides • Animals Spreadsheet • Animals Starter File • The Animals Dataset (Page 2) • Categorical or Quantitative? (Page 3) • Questions and Column Descriptions (Page 4) • *Optional: Opening Questions* • *Optional: What Questions Can You Answer with the Given Data?*
Preparation	• **You know your students best!** You can use the sample Opening Questions we've provided, but we recommend changing or adding your own questions that are *appropriate, relevant, and engaging* for your students. • Decide how the first activity (opening questions) will be run: will questions be printed for each student, group of students, or posted around the room? **Note:** these are just ideas to get you started. Use questions that you know will interest *your* students! • Make sure student computers can access Animals Spreadsheet and the Animals Starter File.
Supplemental Resources	What's going on in this Graph?

Glossary

categorical data :: data whose values are qualities that are not subject to the laws of arithmetic

comments :: messages in the code, generally ignored by the computer, to help people interacting with the code understand what it is doing

data row :: a structured piece of data in a dataset that typically reports all the information gathered about a given individual

data science :: the science of collecting, organizing, and drawing general conclusions from data, with the help of computers

header :: the titles of each column of a table, usually shown at the top

identifier column :: a column of unique values which identify all the individual rows (e.g. - student IDs, SSNs, etc)

programming language :: a set of rules for writing code that a computer can evaluate

quantitative data :: number values for which arithmetic makes sense

Questions that Matter *20 minutes*

Overview

Students look at opening questions, either at their desks or in a walk around the room. They select a question they are personally interested in, and think about the data required to answer that question. This process draws a direct line between answering questions they care about and the basics of data science.

Launch

- Look at the provided list of Opening Questions, and take one minute to select a question that grabs your attention. Arrange yourselves into groups based on the question you like, making sure that each group has between 2-5 people.
- Have each person in the group quickly share their *gut reaction*: What do you think the answer is?
- After sharing, have each person share their reasoning.
- Does everyone in your group agree about the answers to their question?

Note: Students are VERY likely to try and explain their reasoning when they give their gut answers. This can taint the answers of other students in the group - emphasize that this is about exposing our "gut reaction" or bias.

Investigate

What information would you collect to answer the question you selected? *Take 5 minutes to think about what information you would need to collect, to find the answer.*

Common Misconceptions

Students may lean towards questions about *individuals*, instead of questions about what's true for a *group of individuals* who vary from one to another. For example, instead of wondering what movie gets the highest rating, they should ask what's the typical rating for movies in a list, or how much those ratings tend to vary.

Synthesize

For each group…

- What were your gut reactions?
- Did the question wind up being too vague? What did you need to do to make it specific?
- What data would you gather?
- What, if anything, were you surprised about?
  - If we wanted to find out if small schools are better than big schools, for example, we might want to gather data on SAT scores, college acceptance, etc. Each of these is a **variable** in our dataset: any two schools we look at could *vary* by each of them.
  - We can't survey every school in the world (or get data on every movie ever made, or every police action!) but we can analyze a *sample* of them, and try to infer something about all of them as a whole.

These questions quickly turn into a discussion about data — how you assess it, how you interpret the results, and what you can *infer* from those results.

The process of learning from data is called ***Data Science***. Data science techniques are used by scientists, business people, politicians, sports analysts, and hundreds of other different fields to ask and answer questions about data.

We'll use a ***programming language*** to investigate these questions. Just like any human language, programming languages have their own vocabulary and grammar that you will need to learn. The language you'll be learning for data science is called Pyret.

Optional: Which Questions *can* we Answer?

Datasets are useful for answering questions, but they can't answer all the questions that we will wonder about for a given topic. In this activity students will look at a small dataset about a cyclist's training rides and think about how they could use the table to answer each question or why a question cannot be answered using the table.

Which of you like to ride bikes? What data might you collect about bike rides? Open to What Questions Can You Answer with the Given Data? This page includes a small dataset about a cyclist's training rides and a set of questions. The data can be used to answer some, but not all, of the questions. With your partner, read each question. If it can be answered with what we know, explain how you could use the table to answer it. If it can't be answered using the table, explain why not.

Meet the Animals! *25 minutes*

Overview

Students explore the Animals Dataset, sharing observations and familiarizing themselves with the idiosyncrasies and patterns in the data. In the process, they learn about ***Categorical*** and ***Quantitative data***.

Notice and Wonder Pedagogy

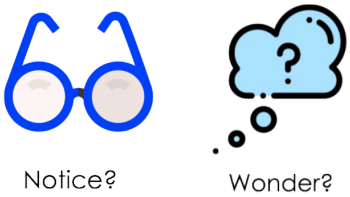
This pedagogy is a widely-used best practice in Math-Ed, and is used throughout this course. In the "Notice" phase, students are asked to crowd-source their observations. No observation is too small or too silly! Students may notice that the animals table has corners, or that it's printed in black ink. But by listening to other students' observations, students may find themselves taking a closer look at the dataset to begin with. The "Wonder" phase involves students raising questions, but they must also explain the context for those questions. Sharon Hessney (moderator for the NYTimes excellent What's going on in this Graph? activity) sometimes calls this "what do you wonder…and **why**?". Both of these phases should be done in groups or as a whole class, with time given to each.

Launch

Open the Animals Spreadsheet in a browser tab, or turn to The Animals Dataset in your workbook.

Investigate

This table contains data from an animal shelter, listing animals that have been adopted. We'll be analyzing this table as an example throughout the course, but you'll be applying what you learn to *a dataset you choose* as well.



- Turn to Questions and Column Descriptions in your Student Workbook. What do you *Notice* about this dataset? Write down your observations in the first column.
- Sometimes, looking at data sparks questions. What do you *Wonder* about this dataset, and why? Write down your questions in the second column.
- There's a third column, called "Answered by Dataset" — circle "Yes" if your Wonder can be answered by the dataset or "No" if it can't.

Have students share back their noticings (statements) and wonderings (questions), and write them on the board. Ask the class if each Wonder can be answered by the data, making sure that they have a few questions that *can* be answered, and a few that *can't*.

- If you look at the bottom of the Animals Spreadsheet, you'll see that this document contains multiple sheets. One is called `"pets"` and the other is called `"README"`. Which sheet are we looking at?

- Each sheet contains a table. For our purposes, we only care about the animals table on the `"pets"` sheet.

Any two animals in our dataset may have different ages, weights, etc. Each of these is called a **variable** in the dataset. Data Scientists work with two broad kinds of data: Categorical Data and Quantitative Data. Sometimes it can be tricky to figure out if data is categorical or quantitative, because it depends on *how that data is being used!*

We use ***Categorical Data*** to answer "what kind?", and ***Quantitative Data*** to answer "how much?".

Categorical Data is used to *classify*, not measure. The laws of arithmetic do not make sense when it comes to categorical data.

- "Species" is a categorical variable, because we can ask questions like "which species does Mittens belong to?"
- We couldn't ask if "cat is more than lizard" and it doesn't make sense to "find the average ZIP code" in a list of addresses.

- What are some other categorical variables you see in this table?
  - *Name, Sex, and Fixed*

Quantitative Data is used to measure an amount of something, or to compare two pieces of data to see which is *less or more*.

- "Pounds" is a quantitative variable, because we can talk about whether one animal weighs more than another or ask what the average weight of animals in the shelter is.
- If we want to ask "how much" or "which is most", we're talking about Quantitative Data.

- What are some other quantitative variables in this table?
  - *Age, Legs, Weight, and Time to Adoption*
- Complete Categorical or Quantitative?. Be sure to discuss your answers with your partner or group!

Synthesize

Data Science is all about using a **smaller sample of data to make educated guesses about a larger population**. It's important to remember that tables are only a *sample* of a larger population: this table describes some animals, but obviously it isn't every animal in the world! Still, if we took the average age of the animals from this particular shelter, it might tell us something about the average age of animals from other shelters.

Meet Pyret! *10 minutes*

Overview

Students open up the Pyret environment (code.pyret.org, or "CPO") and see the Animals Dataset reflected there.

Launch

Let's take a look at our programming environment, and see what the Animals Dataset looks like there.

- Open the Animals Starter File in a new tab. Click "Connect to Google Drive" to sign into your Google account. This will allow you to save Pyret files into your Google Drive.
- Next, click the "File" menu and select "Save a Copy". This will save a copy of the file into your own account, so that you can make changes and retrieve them later.

Investigate

- Click "Run" to tell Pyret to read the code on the left-hand side.
- On the right-hand side, type `animals-table` and hit the "Enter" or "Return" key.
- What happens?
  - *A table appears of animals appears on the right-hand side of the screen.*
- Look on the left-hand side of the screen. Where is Pyret getting `animals-table` from?

The first line on the left-hand side of the screen tells Pyret to use a provided file, which contain tools we'll want to use for this course.

```
use context shared-gdrive("Bootstrap-DataScience-…")
```

After that, we see a line of code that *defines* `shelter-sheet` to be a spreadsheet.

```
shelter-sheet = load-spreadsheet("https://docs.google.com….")
```

This table is loaded from Google Drive, so now Pyret can see the same spreadsheet you do. If you look carefully, you'll notice that the address listed here is the same address as the spreadsheet we just looked at!

What Happens when Pyret loads a spreadsheet?

- When using a spreadsheet, we can make a pie-chart out of any collection of cells - even if those cells are blank, contain mismatched data, or contain data from multiple, unrelated tables! This can jeopardize our analysis, so each cell has to be checked by hand in order for any data displays to be trusted!
- When Pyret imports a sheet, it reads all the cells and builds something it calls a *Table* . The original spreadsheet is still out there, exactly as it used to be! And when building that Table, Pyret does *all of the cell checks automatically* , and holds us accountable for using clean and complete data so that all displays can be trusted.
- Each time we sort, filter or change a Table, Pyret makes a *new Table* . This means that we explore one transformation or another, or even multiple transformations together...and can always go back as many steps as we want.
- When working with spreadsheets we have to choose between making "backup" copies of sheets at every step, or modifying the original sheet. Modifying the sheet makes it really hard to undo. And making copies means the same data is duplicated in many places, so a change in one sheet needs to be replicated *by hand* in other sheets.

This kind of tedious busywork - or the inability to undo a mistake - is problematic for students, teachers, and Data Scientists alike!
For younger students, the strictness of programming may outweigh the benefits. But for students who are comfortable writing code, programming is a vastly more powerful and learner-friendly tool for Data Science!

After that, we see the following code:

```
# load the 'pets' sheet as a table called animals-table
animals-table = load-table: name, species, sex, age, fixed, legs, pounds, weeks
  source: pets-sheet.sheet-by-name("pets", true)
end
```

The first line (starting with `#`) is called a ***Comment***. Comments are notes for humans, which the computer ignores. The next line defines a new table called animals-table, (loaded from the shelter-sheet!) and defines column names: `name` , `species` , `sex` , `age` , `fixed` , `legs` , `pounds` and `weeks` . We could use any names we want for these columns, but it's always a good idea to pick names that make sense!

Even if your spreadsheet already has column headers, Pyret requires that you name them in the program itself.

Every table is made of cells, which are arranged in a grid of rows and columns. *The first row and first column* are special. The first row is called the ***header row***, which gives a unique name to each variable (or "column") in the table. The first column in the table is the ***identifier column***, which contains a unique ID for each row. Often, this will be the name of each individual in the table, or sometimes just an ID number.
Below is an example of a table with one header row and two data rows:

name	species	sex	age	fixed	legs	pounds	weeks
"Sasha"	"cat"	"female"	1	false	4	6.5	3
"Mittens"	"cat"	"female"	2	true	4	7.4	1

- How many variables are listed in the header row for the Animals Dataset? What are they called? What is being used for the identifier column in this dataset?
  - *There are eight variables listed: name, species, sex, age, fixed, legs, pounds, and weeks.* `name` *is the identifier column.*
- Try changing the name of one of the columns, and click "Run". What happens when you look at the table?
  - *The column name also gets changed on the table.*
- What happens if you remove a column from the list? Or add an extra one?
  - *When I remove a column, I get this message: "Loaded worksheet has 8 columns, but 7 column names were given." When I add a column, I get this message: "Loaded worksheet has 8 columns, but 9 column names were given."*

After the header, Pyret tables can have any number of ***data rows***. Each data row has values for every column variable (nothing can be left empty!). A table can have any number of data rows, including *zero* , as in the table below:

name	species	sex	age	fixed	legs	pounds	weeks

Pyret lets us use many different kinds of data. In the animals table, for example, there are Numbers (the number of legs each animal has), Strings (the species of the animal), and Booleans (whether it is true or false that an animal is fixed).

Synthesize

Once you know how to program, you can do a *lot* with datasets:

- Data Scientists **display** tables as all kinds of charts and graphs. For example, we might want to make a pie chart showing how many animals of each species we have.
- Sometimes they want to **filter** a table, showing only a few of the rows. For example we might only want to look at animals where `species` is equal to `"dog"` .
- Or perhaps we want to **build** a column! For example, there could be a vaccination for all cats under the age of 3, and we want to add a `needs-vaccine` column that says `true` for cats under the age of 3, and `false` for everyone else.

In this course, you'll learn how to do all three: Display, Filter, and Build.